

Load Interconnection Request System Impact Study Report

For Queue Project Q14-0387

Gardner 230 kV
October 02, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0387. The Interconnection Customer proposes a 325.6 MW large load (data center) facility interconnecting at the Gardner 230 kV substation in the Illinois area. The request was received on September 28, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 248 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$3,813,000.

3. Network Impacts

3.1 Load Deliverability Analysis

For the loss of Gavin Power Llc 500 kV/345 kV Transformer Bank, the International Paper – Flatlick 500 kV loads to 104.4% of its 3,342 MVA rating. Project DFAX: 5.4%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
International Paper – Flatlick 500 kV	Reconductor line	\$56,440,000	7% (\$3,813,000)
	Total allocated to project		\$3,813,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.